

BdMO 2026 National Camp F.Eqns PSet

M. ZAMIUL HOSSAIN & ABRAR ZAHIN PATHAN

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§1 In-Lecture Problems

Problem 1.1. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$f(x^2 + y) = f(x^{27} + 2y) + f(x^4)$$

for all $x, y \in \mathbb{R}$.

Problem 1.2. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$f(x + yf(x)) + x = f(xf(y)) + f(y + f(x))$$

for all $x, y \in \mathbb{R}$.

Problem 1.3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$f(xf(y) + x) = xy + f(x).$$

Problem 1.4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(f(x) + y) + y) = x + y + f(y)$$

Problem 1.5. Solve over $\mathbb{Z}$ the functional equation

$$f(2a) + 2f(b) = f(f(a + b)).$$

Problem 1.6. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x^2 + y) = f(f(x) - y) + 4f(x)y$$

for all real numbers x and y .

Problem 1.7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ for which

$$f(xf(x) + f(y)) = f(x)^2 + y$$

holds for all real numbers x and y .

§2 Buffer Problems

Problem 2.1. Solve over $\mathbb{R}$ the functional equation

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y).$$

Problem 2.2. Find all strictly increasing functions $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{> 0}$ satisfying $f(2) = 7$ and

$$f(mn) = f(m)f(n) + f(m) + f(n)$$

for all non-negative integers m and n .

Problem 2.3. Find all $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for all **distinct** positive real numbers x, y, z , we have

$$f(x)^2 - f(y)f(z) = f(x^y)f(y)f(z)[f(y^z) - f(z^x)].$$

Problem 2.4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x+y)) = f(yf(x)) + x^2.$$

§3 Resources

- [Introduction to Functional Equations](#)
- [Monster Functional Equations](#)
- [100 Functional Equations - Amir Hossein Parvadi](#)
- [Functional Equations - Pang-Cheng Wu](#)